A Counterexample to the Group-Ring Determinant Conjecture

Rewritten for readability from the OpenAI manuscript of the same title

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Abstract

Taking as input one cited theorem (a group algebra over a finite field of characteristic two that is not directly finite), we construct a finitely generated group G_D and a square matrix A over $\mathbb{Z}[G_D]$ that is invertible over $\mathbb{Q}[G_D]$ and has Fuglede–Kadison determinant strictly between 0 and 1. The logarithmic integral defining the determinant is finite. This contradicts the unrestricted group-ring Determinant Conjecture. Everything after the cited input is proved here in full.

1 The result, its input, and the plan

A nonsingular integer matrix has determinant of absolute value at least one. The group-ring Determinant Conjecture asks for the analogue when the entries are finite integer combinations of elements of a discrete group; the determinant is the one of Fuglede and Kadison [4], defined through the trace of a logarithm.

1.1 Conventions

Throughout, G is a discrete group.

Group rings. For a commutative ring k, $k[G]$ is the ring of *finite* formal sums $\alpha = \sum_{g\in G} \alpha_g\, g$ with $\alpha_g \in k$. We write $[1]\alpha := \alpha_1$ for the coefficient of the identity element. For $k \subseteq \mathbb{C}$ the involution is $\alpha^* = \sum_g \overline{\alpha_g}\, g^{-1}$, and for a matrix A over $k[G]$ we put $(A^*)_{ij} = (A_{ji})^*$; then $(AB)^* = B^*A^*$. A square matrix P over $\mathbb{Q}[G]$ with $P = P^* = P^2$ is called an *orthogonal projection*.

Operators. Let $(\delta_g)_{g\in G}$ be the standard orthonormal basis of $\ell^2(G)$. The left regular representation is $\lambda_g\delta_h = \delta_{gh}$, that is, $(\lambda_g\xi)(h) = \xi(g^{-1}h)$. For $\alpha \in \mathbb{C}[G]$ put $\lambda(\alpha) = \sum_g \alpha_g\lambda_g$. For a matrix $A \in \mathrm{M}_{n'\times n}(\mathbb{C}[G])$ define the bounded operator

$$T_A : \ell^2(G)^n \to \ell^2(G)^{n'}, \qquad (T_A\xi)_i = \sum_{j=1}^{n} \lambda(A_{ij})\,\xi_j, \tag{1.1}$$

so A acts on *column* vectors through the *left* regular representation. Since $\lambda_g\lambda_h = \lambda_{gh}$ and $\lambda_g^* = \lambda_{g^{-1}}$,

$$T_{AB} = T_AT_B, \qquad T_{A^*} = T_A^*, \qquad T_{I_n} = I. \tag{1.2}$$

Two consequences are used constantly. First, if $A \in \mathrm{M}_n(\mathbb{C}[G])$ has an inverse $A^{-1} \in \mathrm{M}_n(\mathbb{C}[G])$, then T_A is invertible with bounded inverse $T_{A^{-1}}$. Second, if P is an orthogonal projection in the sense above, then T_P is an orthogonal projection operator. We also record

$$\lambda(\alpha)\delta_1 = \sum_g \alpha_g\delta_g, \qquad \text{hence} \qquad \langle\lambda(\alpha)\delta_1, \delta_1\rangle = [1]\alpha. \tag{1.3}$$

Trace. Let $\epsilon_i \in \ell^2(G)^n$ be the vector with δ_1 in component i and 0 elsewhere. For a bounded operator T on $\ell^2(G)^n$ put

$$\tau_n(T) = \sum_{i=1}^{n} \langle T\epsilon_i, \epsilon_i \rangle.$$

This trace is *unnormalized*: $\tau_n(I) = n$. For $A \in \mathrm{M}_n(\mathbb{C}[G])$ we write $\tau_n(A) := \tau_n(T_A)$; by (1.1) and (1.3),

$$\tau_n(A) = \sum_{i=1}^{n} [1] A_{ii}. \tag{1.4}$$

Spectral measure and determinant. For $A \in \mathrm{M}_n(\mathbb{Z}[G])$ the operator $T_A^* T_A$ is positive. Its probability spectral measure is

$$\mu_A(E) = \frac{1}{n} \sum_{i=1}^{n} \langle \mathbf{1}_E(T_A^* T_A)\epsilon_i, \epsilon_i \rangle \qquad (E \subseteq [0,\infty) \text{ Borel}).$$

By the spectral theorem, $\tau_n(f(T_A^* T_A)) = n \int f \, d\mu_A$ for every bounded Borel function f on the spectrum of $T_A^* T_A$. If T_A is invertible, then $\|T_A^{-1}\|^{-2} I \le T_A^* T_A \le \|T_A\|^2 I$, so log is bounded on the spectrum, and the determinant convention is

$$\log \det_{\mathcal{N}(G)}(T_A) := \frac{1}{2}\, \tau_n\big(\log(T_A^* T_A)\big) = \frac{n}{2} \int_{(0,\infty)} \log t \, d\mu_A(t). \tag{1.5}$$

In other words $\det_{\mathcal{N}(G)}(T_A) = \exp \tau_n(\log |T_A|)$ with $|T| = (T^*T)^{1/2}$: the Fuglede–Kadison determinant taken with the *unnormalized* trace τ_n. With the normalized trace τ_n/n one would obtain its n-th root; whether the value is < 1 is the same for both. The symbol $\mathcal{N}(G)$, the group von Neumann algebra, only decorates the determinant; no property of it is used. For the trivial group, (1.5) gives $\frac{1}{2} \log \det(A^* A) = \log |\det A|$, the absolute value of the ordinary determinant.

1.2 The conjecture and the main theorem

The statement refuted here is the following. *For every discrete group G, every $n \ge 1$ and every $A \in \mathrm{M}_n(\mathbb{Z}[G])$,*

$$\int_{(0,\infty)} \log t \, d\mu_A(t) \ge 0. \tag{1.6}$$

How to read (1.6). For every A the measure μ_A is carried by $[0, \|T_A\|^2]$, so $\log t$ is bounded above on its support and the integral over the open half-line is a well-defined element of $[-\infty, \infty)$. A possible atom of μ_A at 0 is omitted. For singular A the value may be $-\infty$; (1.6) asserts in particular that this does not happen. When T_A is invertible, (1.5) shows that (1.6) is equivalent to $\det_{\mathcal{N}(G)}(T_A) \ge 1$. The counterexample below is of this invertible kind, so only this case matters for the refutation.

Relation to the literature. "Unrestricted" means that no hypothesis is placed on G: no torsion-freeness, amenability, soficity or anything else. (The group G_D below has elements of order 3.) The source describes (1.6) as the square-matrix instance of [7, Conjecture 13.2], which also permits rectangular matrices, so that a square counterexample already contradicts it. The source does not reproduce the wording of that conjecture, so it is not reproduced here either. Consequently, Theorem 1.1 as a statement about (1.6) is proved in this document (relative to Theorem 1.2). Its last

sentence, "the conjecture has a negative resolution", relies in addition on the source's identification of (1.6) with the cited conjecture, which a reader must compare with the reference. No proof below depends on that identification. Remark 6.1 compares the convention (1.1) with right multiplication on row vectors.

Theorem 1.1. *There are a finitely generated discrete group G_D, an integer $n \geq 1$, and $A \in \mathrm{M}_n(\mathbb{Z}[G_D])$ invertible over $\mathbb{Q}[G_D]$ such that*

$$0 < \det_{\mathcal{N}(G_D)}(T_A) < 1.$$

The integral in Equation (1.6) is finite and strictly negative. Thus the unrestricted group-ring Determinant Conjecture has a negative resolution.

The proof (Section 5) gives $n = 2d + 1$ and $\det_{\mathcal{N}(G_D)}(T_A) = q^{-\theta}$. Here d and θ come out of the construction, and q is any chosen power of two $2^k > 1$ with $q \equiv 1 \pmod{D}$; the matrix A depends on this choice. All three are introduced in Section 1.4.

1.3 The external input

A unital ring is *directly finite* if $ab = 1$ implies $ba = 1$. The construction starts from a failure of direct finiteness, quoted from a companion paper.

Theorem 1.2 ([8, Theorem 1.1])**.** *There exist a finite field K_2 of characteristic two, a finitely generated group H_2, and finite sums $a_2, b_2 \in K_2[H_2]$ with $a_2 b_2 = 1$ and $b_2 a_2 \neq 1$.*

Unverified step. Theorem 1.2 is taken by citation from the companion paper [8] and is not proved or tested in the source or in this rewrite. It is the only unproved step, and the whole result rests on it. It asserts a group algebra over a finite field that is not directly finite, that is, a counterexample to Kaplansky's direct finiteness conjecture; it would also force H_2 to be non-sofic. The source says that it uses the existence theorem, not an executed numerical instance or a group-word decision procedure, so nothing in either document lets a reader check this statement. Everything after this input (Lemma 2.1 through the proof of Theorem 1.1) is proved step by step below, so Theorem 1.1 is established only conditionally on Theorem 1.2.

Remark 1.3 (Status of the input). Theorem 1.2 is the only statement on which Theorem 1.1 depends that is used without proof. It says that the group algebra $K_2[H_2]$ is not directly finite. The displayed statement is all that is used: no further property of K_2, H_2, a_2, b_2 enters. Neither the source nor this rewrite proves or tests it, so Theorem 1.1 is established here *relative to* Theorem 1.2. A reader who wants to certify Theorem 1.1 must check the companion separately.

The source records the following about the companion. Its Section 6 specifies the field, a group presentation and the two sums by a terminating finite prescription. Only the existence statement is used here, not an executed numerical instance or a group-word decision procedure. The field K_2 is finite but is not asserted to be $\mathbb{F}_2$. There is no torsion-free hypothesis, and the companion's construction uses torsion.

Characteristic two is essential below. Lemma 2.1 converts the input into matrices over $\mathbb{F}_2[H_2]$, and Sections 2–3 reduce modulo 2 and modulo powers of 2. A failure of direct finiteness in odd characteristic only would not supply this premise.

1.4 Proof at a glance

Key idea. Suppose $P \in \mathrm{M}_m(\mathbb{Q}[G])$ is an orthogonal projection with odd denominators whose reduction modulo 2 factors as $\bar{P} = uv$ through r columns. Then one can write down an *integral* matrix A with

$$\log \det_{\mathcal{N}(G)}(T_A) = (r - \tau_m(P)) \log q, \qquad q \text{ a suitable power of 2.}$$

So a projection whose real trace exceeds the number of columns of a mod-2 factorization gives determinant < 1.

Sanity check (not used later). For the trivial group, $\tau_m(P) > r$ cannot happen. Let $P \in \mathrm{M}_m(\mathbb{Q})$ be idempotent with entries in the ring $\mathbb{Z}_{(2)}$ of rationals with odd denominator. Then $\mathbb{Z}_{(2)}^m = P\mathbb{Z}_{(2)}^m \oplus (I-P)\mathbb{Z}_{(2)}^m$. Both summands are free, being submodules of a free module over a principal ideal domain; their ranks are a and $m-a$, where a is the rank of P over $\mathbb{Q}$. Since P is diagonalizable with eigenvalues $0, 1$, a equals the trace of P. Reducing modulo 2 gives $\mathbb{F}_2^m = \bar{P}\mathbb{F}_2^m \oplus (I-\bar{P})\mathbb{F}_2^m$ with dimensions a and $m-a$, so $\bar{P}$ has rank a. A factorization $\bar{P} = uv$ through r columns then forces $\tau_m(P) = a \le r$.

Orientation only. Over a group ring the inequality can be reversed, and a failure of direct finiteness over $\mathbb{F}_2$ is what makes this possible in the construction. In Proposition 2.2, Step 6, the nonzero defect $C = I_d - YX$ supplies one extra diagonal entry $\bar{p}$ of the product uv without an extra column.

Steps. Put $d = [K_2 : \mathbb{F}_2]$.

(1) *Binary defect* (Lemma 2.1). Writing multiplication by elements of K_2 as $d \times d$ binary matrices turns a_2, b_2 into $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$ with $XY = I_d$ and a nonzero idempotent $C = I_d - YX$ satisfying $XC = 0$, $CY = 0$.

(2) *Trace gap* (Proposition 2.2). Enlarge the group to $G_D = (\bigoplus_{h \in H_2} C_3) \rtimes H_2$. Pick one nonzero coefficient of one entry of C. With p_h the averaging idempotent of the copy of C_3 labelled h, the element $p = p_1 \prod_{s \in S}(1 - p_s)$ is a rational projection that isolates that coefficient. The result is $P = \mathrm{diag}(I_d, p)$ of size $d+1$ with $\tau_{d+1}(P) = d + \theta$, $\theta > 0$, and a factorization $\bar{P} = uv$ through only d columns over $\mathbb{F}_2[G_D]$. The denominators of P divide $D = 3^{1+|S|}$.

(3) *Lifting* (Lemma 3.1). For every $q = 2^k$ the factorization lifts to $P_q = u_q v_q$ over $(\mathbb{Z}/q\mathbb{Z})[G]$, by a finite geometric series.

(4) *Integral matrix* (Lemma 3.2). Choose $q \equiv 1 \pmod{D}$ and integral lifts U, V of u_q, v_q. With $J = -q^{-1}P - (I_m - P)$ and $B = \mathrm{diag}(qI_r, J)$, the matrix $A = L_1 B R_1$ is integral, where L_x, R_x are block shears with off-diagonal blocks xU/q, xV/q. It has an explicit inverse over $\mathbb{Q}[G]$.

(5) *Shears do not change the determinant* (Lemmas 4.1, 4.2, 4.3). The derivative of $\tau_n(\log(M^*M))$ along a path $M(x)$ is $2 \operatorname{Re} \tau_n(M' M^{-1})$, and this vanishes for $M(x) = L_x B R_x$.

(6) *Evaluation* (Proposition 4.4). At $x = 0$ the matrix is B, for which $\log(B^*B) = 2 \log q \, \mathrm{diag}(I_r, -P)$. Hence $\log \det(T_A) = (r - \tau_m(P)) \log q$.

(7) *Assembly* (Section 5). With $r = d$, $m = d+1$, $\tau_m(P) = d + \theta$ we get $\det(T_A) = q^{-\theta} \in (0, 1)$.

Dependencies.

$$\text{Thm. } 1.2 \Rightarrow \text{Lem. } 2.1 \Rightarrow \text{Prop. } 2.2 \Rightarrow \text{Lem. } 3.1 \Rightarrow \text{Lem. } 3.2 \Rightarrow \text{Prop. } 4.4 \Rightarrow \text{Thm. } 1.1,$$
$$\text{Lem. } 4.1 \Rightarrow \text{Lem. } 4.2 \Rightarrow \text{Lem. } 4.3 \Rightarrow \text{Prop. } 4.4.$$

Sections 3 and 4 are valid for an arbitrary group G and an arbitrary projection P with the stated properties. Only Section 2 and the assembly use the specific group G_D.

1.5 Symbols: renamings and look-alikes

To keep one meaning per symbol, a few of the source's letters are changed. The left regular representation is λ_g (source: L_g), because L_x denotes a shear matrix. The basis vectors are ϵ_i (source: e_i), because e denotes the idempotent P_q in Section 3. The shear blocks in Lemma 4.3 are Λ, Γ (source: S, T), because S is a finite subset of H_2. In Lemma 4.2 the constants and the auxiliary operator are $\kappa_0, \kappa_1, \eta, \Omega$ (source: δ, c, ρ, W), because δ_g, ρ and W have other meanings. Generic group-ring matrices in Lemma 4.1 are α, β (source: C, D). The ring $(\mathbb{Z}/q\mathbb{Z})[G]$ is written out (source: $\mathcal{R}$).

Some letters look alike but are distinguished by typeface or subscript. An italic C without subscript is always the defect matrix $I_d - YX$; C_3 is the cyclic group of order three and $\mathbb{C}$ the complex numbers. D is always the odd denominator-clearing integer, and $d = [K_2 : \mathbb{F}_2]$. Italic B is the diagonal matrix $\mathrm{diag}(qI_r, J)$ of Section 3; calligraphic $\mathcal{B}$ is the subgroup $\bigoplus_h C_3$ of G_D.

1.6 Context

This subsection is not used in the proofs. The statements about the literature, including the numerical values quoted from [5], are the source's and are not checked here.

Determinant bounds are part of the theory of L^2-invariants [7]. Approximation and closure arguments established the bound (1.6) for large classes of groups, including amenable groups [9, Theorem 1.21, with its erratum]; Elek and Szabó proved it for sofic groups [3, Theorem 5], using finite graph approximations and the integrality of products of nonzero eigenvalues of positive integral matrices. The present construction starts instead from a failure of direct finiteness in characteristic two. As a separate consequence, the sofic theorem implies that G_D is not sofic (Remark 6.2).

The lower bound $\det \geq 1$ of (1.6) is a different question from determinant *approximation*, which asks whether the determinant over G is the limit of determinants over finite quotients. Kammeyer's counterexample to such a limiting formula has determinant 2 over the discrete Heisenberg group, while over its principal congruence quotients of levels 3^i, $i \geq 1$, the determinants are at most $5^{1/3}$ [5, Theorem 2 and Section 2]. That example shows a failure of convergence. The determinant over the Heisenberg group there is $2 \geq 1$, so the inequality (1.6) is not violated by it.

The projection p of Section 2 is related to the commuting wreath-product projections of Dicks and Schick [2, Section 3] and of Lehner, Neuhauser and Woess [6, Section 2]. No spectral-distribution theorem is imported from those works. The analytic step is the classical tracial logarithmic derivative [4, Lemma 2], also present in the path-integral viewpoint of de la Harpe and Skandalis [1, Lemma 1 and Proposition 2(d)]; it is proved below (Lemma 4.2).

2 From the scalar defect to a projection

This section produces the trace gap: a rational projection of trace $d + \theta$ whose reduction modulo 2 factors through d columns. All group-ring sums are finite, and the order of factors in every product is kept.

Lemma 2.1. *Let K_2 be a finite field of characteristic two and H_2 a group. For $d = [K_2 : \mathbb{F}_2]$, there is an injective unital ring homomorphism*

$$\rho : K_2[H_2] \longrightarrow \mathrm{M}_d(\mathbb{F}_2[H_2]).$$

For $a_2, b_2 \in K_2[H_2]$ with $a_2 b_2 = 1 \neq b_2 a_2$, their images $X = \rho(a_2)$, $Y = \rho(b_2)$ satisfy $XY = I_d$ and $C := I_d - YX \neq 0$, with

$$C^2 = C, \qquad XC = 0, \qquad CY = 0. \tag{2.1}$$

Where it is used. It supplies the matrices X, Y to which Proposition 2.2 is applied.

Proof. **Step 1 (field elements as binary matrices).** Choose an $\mathbb{F}_2$-basis $\beta_1, \dots, \beta_d$ of K_2. For $c \in K_2$ let $\rho_0(c) \in \mathrm{M}_d(\mathbb{F}_2)$ be the matrix of the $\mathbb{F}_2$-linear map $y \mapsto cy$, that is, $c\beta_j = \sum_i \rho_0(c)_{ij}\beta_i$. Composition of maps gives $\rho_0(cc') = \rho_0(c)\rho_0(c')$; also $\rho_0(c + c') = \rho_0(c) + \rho_0(c')$ and $\rho_0(1) = I_d$. If $\rho_0(c) = 0$ then $cy = 0$ for all y, so $c = c \cdot 1 = 0$. Thus ρ_0 is an injective unital ring homomorphism.
Step 2 (extension to the group ring). For $a = \sum_h c_h\, h \in K_2[H_2]$ define $\rho(a) \in \mathrm{M}_d(\mathbb{F}_2[H_2])$ by

$$\rho(a)_{ij} = \sum_h \rho_0(c_h)_{ij}\, h.$$

This map is additive and $\rho(1) = I_d$. For $a = \sum_g c_g\, g$ and $b = \sum_{g'} c'_{g'}\, g'$ we have $ab = \sum_{g,g'} c_g c'_{g'}\, gg'$, and

$$\big(\rho(a)\rho(b)\big)_{ij} = \sum_k \sum_{g,g'} \rho_0(c_g)_{ik}\, \rho_0(c'_{g'})_{kj}\; gg' = \sum_{g,g'} \rho_0(c_g c'_{g'})_{ij}\; gg' = \rho(ab)_{ij}.$$

The first equality uses only that elements of $\mathbb{F}_2$ commute with group elements; the order gg' is never changed, so non-commutativity of H_2 plays no role. The second is $\rho_0(c)\rho_0(c') = \rho_0(cc')$, and the third is additivity of ρ. If $\rho(a) = 0$, then for each h all entries of $\rho_0(c_h)$ vanish, so $c_h = 0$ by Step 1; hence ρ is injective.
Step 3 (the identities). $XY = \rho(a_2 b_2) = \rho(1) = I_d$. Since $b_2 a_2 \neq 1$ and ρ is injective, $YX = \rho(b_2 a_2) \neq I_d$, so $C \neq 0$. The remaining identities use only $XY = I_d$ and hold in any ring; in particular they do not use characteristic two:

$$\begin{aligned} (YX)^2 &= Y(XY)X = YX, \\ C^2 &= I_d - 2\,YX + (YX)^2 = I_d - 2\,YX + YX = I_d - YX = C, \\ XC &= X - (XY)X = 0, \qquad CY = Y - Y(XY) = 0. \end{aligned}$$

□

The group G_D. Let $\mathcal{B} = \bigoplus_{h \in H_2} C_3$ be the direct sum of copies of the cyclic group of order 3, one for each $h \in H_2$, and let z_h be a generator of the copy labelled h. Every element of $\mathcal{B}$ has a unique *normal form* $\prod_h z_h^{\nu_h}$ with exponents $\nu_h \in \{0, 1, 2\}$, almost all zero; the identity is the element with all $\nu_h = 0$. The group H_2 acts on $\mathcal{B}$ by left translation of labels, $h \cdot z_{h'} = z_{hh'}$, and $G_D = \mathcal{B} \rtimes H_2$. Concretely, G_D contains $\mathcal{B}$ as an abelian normal subgroup and H_2 as a subgroup, and

$$h\, z_{h'}\, h^{-1} = z_{hh'} \qquad (h, h' \in H_2). \tag{2.2}$$

Since $H_2 \leq G_D$, we regard $\mathbb{F}_2[H_2] \subseteq \mathbb{F}_2[G_D]$.

Proposition 2.2. *Let $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$, $d \geq 1$, satisfy $XY = I_d \neq YX$. In the group $G_D = (\bigoplus_{h \in H_2} C_3) \rtimes H_2$ just described, there is an orthogonal projection $P \in \mathrm{M}_{d+1}(\mathbb{Z}[1/3][G_D])$ and a finite set $S \subseteq H_2 \setminus \{1\}$ such that*

$$\tau_{d+1}(P) = d + \theta, \qquad \theta = \frac{1}{3}\left(\frac{2}{3}\right)^{|S|} > 0, \qquad \bar{P} = uv,$$

where $\bar{P}$ is the reduction of P modulo 2, and u has size $(d+1)\times d$ and v has size $d\times(d+1)$ over $\mathbb{F}_2[G_D]$. The integer $D=3^{1+|S|}$ clears all coefficients of P, that is, $DP\in \mathrm{M}_{d+1}(\mathbb{Z}[G_D])$.

Where it is used. P, u, v, D are the input of Proposition 4.4 in the proof of Theorem 1.1.
Plan. Put $C=I_d-YX$ and normalize one entry of C to the form $1+\sum_{s\in S}s$ (Step 1). Build a rational projection p with $psp=0$ for all $s\in S$ (Steps 2–3), so that sandwiching between two copies of p kills every term except the 1. Compute its trace (Step 4), reduce modulo 2 (Step 5), and write down u, v (Step 6).

Proof. Put $C=I_d-YX$. Step 3 of the proof of Lemma 2.1 used only $XY=I_d\neq YX$, so $C\neq 0$ and the identities (2.1) hold.

Step 1 (normalizing one entry). Since $C\neq 0$, choose indices i,j and $h_0\in H_2$ such that the coefficient of h_0 in $C_{ij}\in\mathbb{F}_2[H_2]$ is nonzero. It equals 1, because the coefficient field is $\mathbb{F}_2$. Left multiplication by h_0^{-1} moves this term to the identity:

$$h_0^{-1}C_{ij}=1+\sum_{s\in S}s, \tag{2.3}$$

where $S=\{h_0^{-1}g:\ g\neq h_0 \text{ in the support of } C_{ij}\}$ is a finite subset of $H_2\setminus\{1\}$, possibly empty.

Step 2 (the projection p). For $h\in H_2$ put

$$p_h=\frac{1+z_h+z_h^2}{3}\in\mathbb{Q}[\mathcal{B}],\qquad p=p_1\prod_{s\in S}(1-p_s).$$

We claim $p=p^*=p^2$ and $3^{1+|S|}p\in\mathbb{Z}[G_D]$.

- $p_h^*=p_h$: inversion permutes $\{1,z_h,z_h^2\}$.
- $p_h^2=p_h$: the nine products in $(1+z_h+z_h^2)^2$ give each element of $\{1,z_h,z_h^2\}$ three times, so $(1+z_h+z_h^2)^2=3(1+z_h+z_h^2)$.
- Hence each $1-p_s$ is also a self-adjoint idempotent.
- All these elements lie in the commutative ring $\mathbb{Q}[\mathcal{B}]$. A product of commuting self-adjoint idempotents is a self-adjoint idempotent, so $p=p^*=p^2$.
- $3^{1+|S|}p=(1+z_1+z_1^2)\prod_{s\in S}(2-z_s-z_s^2)$ has integer coefficients.

This is the elementary commuting-coordinate method of the works cited in Section 1; compare [2, Lemmas 3.1 and 3.4].

Step 3 (separation: $psp=0$ for $s\in S$). By (2.2), $s\,p_h\,s^{-1}=p_{sh}$ for $s,h\in H_2$. Conjugation by s is a ring automorphism of $\mathbb{Q}[G_D]$, so

$$s\,p\,s^{-1}=p_s\prod_{t\in S}(1-p_{st})\in\mathbb{Q}[\mathcal{B}].$$

Fix $s\in S$. Then p contains the factor $1-p_s$ and sps^{-1} contains the factor p_s. All factors commute and $(1-p_s)p_s=p_s-p_s^2=0$, so $p\,(sps^{-1})=0$. Therefore

$$p\,s\,p=p\,(sps^{-1})\,s=0\qquad(s\in S). \tag{2.4}$$

Only this single pair of factors is used. The label sets $\{1\}\cup S$ of p and $\{s\}\cup sS$ of sps^{-1} may overlap in other places, for instance when $s^{-1}\in S$ or s has finite order, without affecting the argument.

Step 4 (the trace of p). We have $p_1 = \frac{1}{3}(1 + z_1 + z_1^2)$ and $1 - p_s = \frac{2}{3} - \frac{1}{3}z_s - \frac{1}{3}z_s^2$. The labels in $\{1\} \cup S$ are pairwise distinct, because $1 \notin S$ and S is a set. Expand the product p by choosing one term from each factor: a scalar times $z_1^{\nu_1}$ from p_1 and a scalar times $z_s^{\nu_s}$ from each $1 - p_s$, with all exponents in $\{0, 1, 2\}$. Because the labels are distinct, the product of the chosen group elements is $z_1^{\nu_1} \prod_{s \in S} z_s^{\nu_s}$, which is already in normal form; no two exponents are added. By uniqueness of the normal form it is the identity only if all exponents are 0. So the only term of the expansion that contributes to the identity coefficient is the product of the constant terms of the factors, and by (1.4)

$$\tau_1(p) = [1]p = \frac{1}{3} \prod_{s \in S} \frac{2}{3} = \theta > 0. \tag{2.5}$$

For empty S this reads $p = p_1$, $\theta = 1/3$.

Step 5 (reduction modulo two). Since 3 is odd, there is a ring homomorphism $\mathbb{Z}[1/3] \to \mathbb{F}_2$ (sending 3 and $1/3$ to 1). Applied to coefficients, it gives a ring homomorphism $\alpha \mapsto \bar{\alpha}$ from $\mathbb{Z}[1/3][G_D]$ to $\mathbb{F}_2[G_D]$ that fixes group elements. Hence $\bar{p}^2 = \bar{p}$ and, from (2.4), $\bar{p}\, s\, \bar{p} = 0$ for $s \in S$. Multiplying (2.3) by $\bar{p}$ on both sides,

$$\bar{p}\, h_0^{-1} C_{ij}\, \bar{p} = \bar{p} \cdot 1 \cdot \bar{p} + \sum_{s \in S} \bar{p}\, s\, \bar{p} = \bar{p}. \tag{2.6}$$

Step 6 (the factorization). Let f be the row of length d with entry $\bar{p}\, h_0^{-1}$ at position i and zero elsewhere, and let w be the column of length d with entry $\bar{p}$ at position j and zero elsewhere. Then $fCw = \bar{p}\, h_0^{-1} C_{ij}\, \bar{p} = \bar{p}$ by (2.6). Define

$$P = \operatorname{diag}(I_d, p), \qquad u = \begin{pmatrix} X \\ fC \end{pmatrix}, \qquad v = \begin{pmatrix} Y & Cw \end{pmatrix},$$

of sizes $(d+1) \times (d+1)$, $(d+1) \times d$ and $d \times (d+1)$. Using $XY = I_d$ and (2.1) ($XC = 0$, $CY = 0$, $C^2 = C$),

$$uv = \begin{pmatrix} XY & XCw \\ fCY & fC^2w \end{pmatrix} = \begin{pmatrix} I_d & 0 \\ 0 & fCw \end{pmatrix} = \operatorname{diag}(I_d, \bar{p}) = \bar{P}.$$

Step 7 (conclusion). By Step 2, $P = P^* = P^2$, its coefficients lie in $\mathbb{Z}[1/3]$, and $D = 3^{1+|S|}$ clears them. By (1.4) and (2.5), $\tau_{d+1}(P) = d + [1]p = d + \theta$. It is the rational projection P, not its binary reduction, that carries this trace. □

3 Lifting the factorization and preserving integrality

This section and the next work with the following general data.

Standing data. G is any group; $m, r \geq 1$; $P \in \mathrm{M}_m(\mathbb{Q}[G])$ satisfies $P^2 = P$; D is an odd positive integer with $DP \in \mathrm{M}_m(\mathbb{Z}[G])$; and $\bar{P} = uv$ with $u \in \mathrm{M}_{m \times r}(\mathbb{F}_2[G])$, $v \in \mathrm{M}_{r \times m}(\mathbb{F}_2[G])$.

Reductions. The coefficients of P lie in $\mathbb{Z}[1/D]$. For $q = 2^k$, $k \geq 1$, the odd integer D is a unit modulo q, so there is a unique ring homomorphism $\pi_q : \mathbb{Z}[1/D] \to \mathbb{Z}/q\mathbb{Z}$; on $\mathbb{Z}$ it is the usual reduction. Applying π_q to coefficients gives ring homomorphisms on group rings and on matrices. We write $P_q = \pi_q(P)$ and $\bar{P} = P_2$. By uniqueness, π_2 is π_q followed by reduction $\mathbb{Z}/q\mathbb{Z} \to \mathbb{F}_2$; so $\bar{P}$ is the reduction modulo 2 of P_q.

Motivation. Let $q > 1$ be a power of two. The diagonal operator that multiplies r coordinate summands by q and the range of P by q^{-1} has logarithmic determinant $(r - \tau_m(P)) \log q$, which is negative when $\tau_m(P) > r$. It is not integral. We make it integral by adding block shears whose off-diagonal blocks come from a factorization of P modulo q. Section 4 shows that the shears do not change the determinant.

Lemma 3.1. *Let G be a group, $m, r \geq 1$, D an odd positive integer, and $P \in \mathrm{M}_m(\mathbb{Q}[G])$ satisfy $P^2 = P$ and $DP \in \mathrm{M}_m(\mathbb{Z}[G])$. Suppose $\bar{P} = uv$ over $\mathbb{F}_2[G]$, where u and v have respective sizes $m \times r$ and $r \times m$. For $q = 2^k$, $k \geq 1$, the reduction $e = P_q$ has a factorization $e = u_q v_q$ over $(\mathbb{Z}/q\mathbb{Z})[G]$, with u_q of size $m \times r$ and v_q of size $r \times m$.*

Where it is used. Its factors u_q, v_q are lifted to integral matrices U, V in Lemma 3.2. Of the properties of a projection, only $P^2 = P$ is used here, not $P^* = P$; the oddness of D is used to define the reductions.
Plan. Lift u, v arbitrarily to u_0, v_0 modulo q. Then eu_0v_0e differs from e by a matrix with even coefficients, which is nilpotent modulo 2^k; invert by a finite geometric series.

Proof. **Step 1.** $e^2 = e$, because π_q is a ring homomorphism and $P^2 = P$.

Step 2. Since $\mathbb{Z}/q\mathbb{Z} \to \mathbb{F}_2$ is surjective, we may choose (coefficient by coefficient) matrices u_0, v_0 over $(\mathbb{Z}/q\mathbb{Z})[G]$ of sizes $m \times r$, $r \times m$ whose reductions modulo 2 are u, v. Put

$$Z = e\, u_0 v_0\, e, \qquad W = e - Z.$$

By Step 1, $eZ = Z = Ze$ and $eW = W = We$. So e acts as an identity element on Z and W (they lie in the "corner ring" $e\,\mathrm{M}_m((\mathbb{Z}/q\mathbb{Z})[G])e$, whose identity is e).

Step 3 ($W^k = 0$). Modulo 2, Z reduces to $\bar{P}\,uv\,\bar{P} = \bar{P}^3 = \bar{P}$ and e reduces to $\bar{P}$. So W reduces to 0: every coefficient of W is even, and $W = 2W_1$ for some matrix W_1 over $(\mathbb{Z}/q\mathbb{Z})[G]$. As 2 is central, $W^k = 2^k W_1^k = qW_1^k = 0$. This is a statement about the single matrix W; nothing is claimed about other elements of the corner ring.

Step 4 (an inverse of Z relative to e). Put $Q = e + W + \cdots + W^{k-1}$; for $k = 1$ this is $Q = e$. Using only $Z = e - W$, $We = eW = W$ and $W^k = 0$,

$$QZ = (e + W + \cdots + W^{k-1}) - (W + W^2 + \cdots + W^k) = e - W^k = e,$$

and likewise $ZQ = e$. The claim is $QZ = ZQ = e$, not $QZ = I_m$: Z need not be invertible in the full matrix ring.

Step 5. Set $u_q = Q\, e\, u_0$ and $v_q = v_0\, e$. Then $u_q v_q = Q\,(e\, u_0 v_0\, e) = QZ = e$. Only finite sums and products were used, so all supports are finite. □

The construction. In addition to the standing data, assume $P^* = P$. Fix:

- $q = 2^k > 1$ with $q \equiv 1 \pmod{D}$. Such k exists: 2 is a unit modulo the odd integer D, so $2^k \equiv 1 \pmod{D}$ for k the multiplicative order of 2 modulo D; for $D = 1$ any $k \geq 1$ works.
- u_q, v_q over $(\mathbb{Z}/q\mathbb{Z})[G]$ with $u_q v_q = e = P_q$, from Lemma 3.1.
- $U \in \mathrm{M}_{m\times r}(\mathbb{Z}[G])$ and $V \in \mathrm{M}_{r\times m}(\mathbb{Z}[G])$, arbitrary matrices whose reductions modulo q are u_q and v_q. Consequently

$$UV \equiv u_q v_q = e \pmod{q}. \tag{3.1}$$

- $J = -q^{-1}P - (I_m - P) \in \mathrm{M}_m(\mathbb{Q}[G])$ and $B = \mathrm{diag}(qI_r, J)$.
- For real x, the block shears

$$L_x = \begin{pmatrix} I_r & 0 \\ xU/q & I_m \end{pmatrix}, \qquad R_x = \begin{pmatrix} I_r & xV/q \\ 0 & I_m \end{pmatrix}.$$

The choice of J is forced by integrality. If the upper-left block is qI_r and the off-diagonal blocks are V, U, then eliminating the off-diagonal blocks subtracts $U(qI_r)^{-1}V = UV/q$ from the lower-right block. For the Schur complement to be J, the lower-right block must be $UV/q+J = q^{-1}(UV-P-q(I_m-P))$. The minus signs in J produce the difference $UV - P$, which the congruence (3.1) makes divisible by q. They do not affect J^2.

Lemma 3.2. *Suppose $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$, an odd $D > 0$ clears its denominators, and $\bar{P} = uv$ factors through $r \geq 1$ columns. Let $q = 2^k > 1$ satisfy $q \equiv 1 \pmod{D}$, and let U, V be integral lifts of the factors u_q, v_q supplied by Lemma 3.1, as in the construction above. Then*

$$A = \begin{pmatrix} qI_r & V \\ U & q^{-1}(UV - P - q(I_m - P)) \end{pmatrix} \in \mathrm{M}_{r+m}(\mathbb{Z}[G]) \tag{3.2}$$

has an inverse in $\mathrm{M}_{r+m}(\mathbb{Q}[G])$. More precisely, for every real x,

$$M(x) := L_x B R_x = \begin{pmatrix} qI_r & xV \\ xU & x^2UV/q + J \end{pmatrix}, \qquad M(x)^{-1} = R_{-x}\, B^{-1}\, L_{-x}, \tag{3.3}$$

and $A = M(1)$, $B = M(0)$.

Where it is used. Proposition 4.4 applies Lemma 4.3 to the path $M(x)$, $x \in [0, 1]$.

Proof. **Step 1 (integrality).** The blocks qI_r, U, V are integral. For the last block, write $P + q(I_m - P) = qI_m + (1 - q)P$. This is integral, because $D \mid (q - 1)$ and DP is integral. Apply π_q: since $\pi_q(q) = 0$ and $\pi_q(1 - q) = 1$, its reduction modulo q is $\pi_q(P) = e$. By (3.1), UV also reduces to e. So $UV - P - q(I_m - P)$ is an integral matrix whose reduction modulo q is $e - e = 0$; every coefficient is divisible by q, and the lower-right block of (3.2) is integral.

Step 2 (inverse of B). Since $P^2 = P$, we have $P(I_m - P) = 0 = (I_m - P)P$ and $(I_m - P)^2 = I_m - P$. Hence

$$(-q^{-1}P - (I_m - P))(-qP - (I_m - P)) = P + (I_m - P) = I_m,$$

and the same in the other order. So $J^{-1} = -qP - (I_m - P)$ and $B^{-1} = \mathrm{diag}(q^{-1}I_r, J^{-1})$, both with coefficients in $\mathbb{Q}$.

Step 3 (the product). Multiplying in the given order,

$$L_x B = \begin{pmatrix} qI_r & 0 \\ xU & J \end{pmatrix}, \qquad (L_x B)R_x = \begin{pmatrix} qI_r & xV \\ xU & x^2UV/q + J \end{pmatrix}.$$

At $x = 1$ the lower-right block is $UV/q - q^{-1}P - (I_m - P) = q^{-1}(UV - P - q(I_m - P))$, so $M(1) = A$. At $x = 0$, $L_0 = R_0 = I$ and $M(0) = B$.

Step 4 (inverse). Direct multiplication gives $L_x L_{-x} = I = R_x R_{-x}$, so $M(x)^{-1} = R_{-x} B^{-1} L_{-x}$. For $x = 1$ all coefficients are rational, so $A^{-1} \in \mathrm{M}_{r+m}(\mathbb{Q}[G])$. □

For real x the entries of $M(x)$ and of $M(x)^{-1}$ are polynomials in x with coefficients in $\mathrm{M}(\mathbb{Q}[G])$, hence lie in $\mathrm{M}(\mathbb{R}[G])$, the group ring with real coefficients. The path is integral at $x = 1$; for $0 < x < 1$ it need not be. No relation such as $PU = U$, $VP = V$ or $V = U^*$ is required of the lifts.

Remark 3.3 (Sanity check: trivial group). Let G be trivial, $m = r = 1$, $P = (1)$, $D = 1$, $u = v = (1)$, $q = 2$. Lemma 3.1 gives $u_q = v_q = 1$. With lifts $U = V = 1$, (3.2) is $A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$; with $U = 3$, $V = 1$ it is $A = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$. Both have $|\det A| = 1 = q^{\,r-\tau_1(P)}$, in agreement with Proposition 4.4 below and independent of the lifts.

4 The trace calculation

Let $\mathcal{A}_n$ be the operator-norm closure of $\{T_A : A \in \mathrm{M}_n(\mathbb{C}[G])\}$ in the bounded operators on $\ell^2(G)^n$. By (1.2) it is a unital C^*-algebra. Its elements are exactly the $n \times n$ matrices with entries in $\mathcal{A}_1$. Indeed, if T_{ij} denotes the (i, j) entry of an operator T on $\ell^2(G)^n$ (the i-th component of T restricted to the j-th summand), then $\|T_{ij}\| \le \|T\| \le \sum_{i,j} \|T_{ij}\|$, so a sequence of operator matrices converges in norm if and only if it converges entrywise. From now on a group-ring matrix A is identified with its operator T_A.

Lemma 4.1 (Properties of τ_n)**.** *For $T, T' \in \mathcal{A}_n$:*

(a) *τ_n is linear, $|\tau_n(T)| \le n\|T\|$ and $\tau_n(T^*) = \overline{\tau_n(T)}$;*

(b) *$\tau_n(TT') = \tau_n(T'T)$;*

(c) *$\tau_n(T) = \sum_i \langle T_{ii}\delta_1, \delta_1\rangle$ depends only on the diagonal entries $T_{ii} \in \mathcal{A}_1$ of T; in particular $\tau_n(T) = 0$ if these vanish.*

Where it is used. In Lemmas 4.2 and 4.3.

Proof. (a) and (c) follow from $\tau_n(T) = \sum_i \langle T\epsilon_i, \epsilon_i\rangle$: each ϵ_i is a unit vector, $\langle T^*\epsilon_i, \epsilon_i\rangle = \overline{\langle T\epsilon_i, \epsilon_i\rangle}$, and $\langle T\epsilon_i, \epsilon_i\rangle = \langle T_{ii}\delta_1, \delta_1\rangle$.

(b) First let $T = T_\alpha$, $T' = T_\beta$ with $\alpha, \beta \in \mathrm{M}_n(\mathbb{C}[G])$. By (1.4),

$$\tau_n(\alpha\beta) = \sum_{i,j} [1](\alpha_{ij}\beta_{ji}) = \sum_{i,j}\sum_{g\in G} (\alpha_{ij})_g\,(\beta_{ji})_{g^{-1}}, \qquad \tau_n(\beta\alpha) = \sum_{i,j}\sum_{h\in G} (\beta_{ji})_h\,(\alpha_{ij})_{h^{-1}}.$$

The substitution $h = g^{-1}$ identifies the two sums. For general T, T' choose group-ring matrices with $T_{\alpha_k} \to T$, $T_{\beta_k} \to T'$ in norm; then $T_{\alpha_k}T_{\beta_k} \to TT'$ and $T_{\beta_k}T_{\alpha_k} \to T'T$, and τ_n is norm-continuous by (a). □

A path $M : [0, 1] \to \mathcal{A}_n$ is *continuously differentiable* if it is differentiable in the operator norm with continuous derivative M'.

Lemma 4.2. *For a continuously differentiable path $M : [0, 1] \to \mathcal{A}_n$ of invertible elements,*

$$\frac{d}{dx}\tau_n\big(\log(M(x)^*M(x))\big) = 2\,\mathrm{Re}\,\tau_n\big(M'(x)M(x)^{-1}\big).$$

Where it is used. In Lemma 4.3. This is the tracial logarithmic derivative of [4, Lemma 2], written with the unnormalized trace.
Plan. Put $N = M^*M$. Expand $\log N$ in a power series around a large multiple of the identity, differentiate τ_n of the series term by term using cyclicity, and resum to get $\tau_n(N^{-1}N')$. Then express N' through M'.

Proof. **Step 1 (uniform bounds).** Put $N(x) = M(x)^*M(x)$. Inversion is continuous on the invertible elements of a Banach algebra, so $x \mapsto M(x)^{-1}$ is continuous on $[0,1]$. Hence $\kappa_1 := \sup_x \|M(x)\|^2$ and $\kappa_0 := (\sup_x \|M(x)^{-1}\|)^{-2}$ are finite and positive. For every vector ξ, $\langle N\xi, \xi\rangle = \|M\xi\|^2$ lies between $\|M^{-1}\|^{-2}\|\xi\|^2$ and $\|M\|^2\|\xi\|^2$, so

$$\kappa_0 I \le N(x) \le \kappa_1 I \qquad (x \in [0,1]).$$

Put $\Omega = I - N/\kappa_1$ and $\eta = 1 - \kappa_0/\kappa_1 \in [0,1)$. Then $0 \le \Omega \le \eta I$, so $\|\Omega(x)\| \le \eta < 1$ for all x.
Step 2 (series for the logarithm). Since $N = \kappa_1(I - \Omega)$ and $\log(\kappa_1(1-t)) = \log\kappa_1 - \sum_{j\ge1} t^j/j$ uniformly for $t \in [0,\eta]$, functional calculus gives the norm-convergent series

$$\log N = (\log\kappa_1)I - \sum_{j\ge1} \frac{\Omega^j}{j}.$$

By linearity and continuity of τ_n (Lemma 4.1(a)),

$$\tau_n(\log N) = n\log\kappa_1 - \sum_{j\ge1}\phi_j, \qquad \phi_j(x) := \frac{\tau_n(\Omega(x)^j)}{j}, \qquad |\phi_j| \le \frac{n\eta^j}{j}.$$

Step 3 (termwise derivative). $N = M^*M$ is continuously differentiable with $N' = M'^*M + M^*M'$, where $M'^* := (M')^* = (M^*)'$; so $\Omega' = -N'/\kappa_1$ is continuous. The product rule gives $\frac{d}{dx}\Omega^j = \sum_{i=0}^{j-1} \Omega^i\,\Omega'\,\Omega^{j-1-i}$. Commutation of Ω and Ω' is not assumed. By Lemma 4.1(b), each of the j terms has trace $\tau_n(\Omega^{j-1}\Omega')$. Hence

$$\phi_j' = \tau_n(\Omega^{j-1}\Omega') = -\frac{1}{\kappa_1}\tau_n(\Omega^{j-1}N'), \qquad |\phi_j'| \le n \sup_{y\in[0,1]} \|\Omega'(y)\|\,\eta^{j-1}.$$

The bounds on $|\phi_j|$ (Step 2) and $|\phi_j'|$ are summable and independent of x, so both $\sum\phi_j$ and $\sum\phi_j'$ converge uniformly on $[0,1]$. By the standard theorem on termwise differentiation of series of C^1 functions, the series may be differentiated term by term:

$$\frac{d}{dx}\tau_n(\log N) = \frac{1}{\kappa_1}\sum_{j\ge1}\tau_n(\Omega^{j-1}N') = \frac{1}{\kappa_1}\tau_n((I-\Omega)^{-1}N') = \tau_n(N^{-1}N').$$

The middle equality is the norm-convergent Neumann series $\sum_{j\ge1}\Omega^{j-1} = (I-\Omega)^{-1}$; the last uses $I - \Omega = N/\kappa_1$.
Step 4 (from N to M). Insert $N' = M'^*M + M^*M'$ and $N^{-1} = M^{-1}(M^*)^{-1}$.

- Second summand: $N^{-1}M^*M' = M^{-1}M'$, and $\tau_n(M^{-1}M') = \tau_n(M'M^{-1})$ by cyclicity.
- First summand: by cyclicity $\tau_n(N^{-1}M'^*M) = \tau_n(MN^{-1}M'^*) = \tau_n((M^*)^{-1}M'^*)$, because $MN^{-1} = (M^*)^{-1}$. Since $(M^*)^{-1}M'^* = (M'M^{-1})^*$, Lemma 4.1(a) gives the value $\overline{\tau_n(M'M^{-1})}$.

The sum is $2\,\mathrm{Re}\,\tau_n(M'M^{-1})$. □

Lemma 4.3. *Let $r, m \geq 1$ and let $B \in \mathcal{A}_{r+m}$ be invertible. Let Λ and Γ be matrices of sizes $m \times r$ and $r \times m$ with entries in $\mathcal{A}_1$. For $x \in [0,1]$ put*

$$L_x = \begin{pmatrix} I_r & 0 \\ x\Lambda & I_m \end{pmatrix}, \qquad R_x = \begin{pmatrix} I_r & x\Gamma \\ 0 & I_m \end{pmatrix}.$$

Then $\tau_{r+m}(\log((L_xBR_x)^*(L_xBR_x)))$ is constant in x.

Where it is used. In Proposition 4.4, with $\Lambda = U/q$, $\Gamma = V/q$ and $B = \operatorname{diag}(qI_r, J)$.

Proof. Write $M(x) = L_xBR_x$. It is a polynomial in x with coefficients in $\mathcal{A}_{r+m}$, hence continuously differentiable. As $L_xL_{-x} = I = R_xR_{-x}$, it is invertible with $M^{-1} = R_x^{-1}B^{-1}L_x^{-1}$. The product rule, keeping the order of factors, gives $M' = L_x'BR_x + L_xBR_x'$, so

$$M'M^{-1} = L_x'L_x^{-1} + (L_xB)\,(R_x'R_x^{-1})\,(L_xB)^{-1}.$$

Here

$$L_x'L_x^{-1} = \begin{pmatrix} 0 & 0 \\ \Lambda & 0 \end{pmatrix}\begin{pmatrix} I_r & 0 \\ -x\Lambda & I_m \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ \Lambda & 0 \end{pmatrix}, \qquad R_x'R_x^{-1} = \begin{pmatrix} 0 & \Gamma \\ 0 & 0 \end{pmatrix}.$$

Both have zero diagonal blocks, hence zero diagonal entries, so their traces vanish by Lemma 4.1(c). By cyclicity (Lemma 4.1(b)), the conjugate $(L_xB)(R_x'R_x^{-1})(L_xB)^{-1}$ has the same trace as $R_x'R_x^{-1}$, namely 0. Thus $\tau_{r+m}(M'M^{-1}) = 0$, and Lemma 4.2 gives derivative zero on $[0,1]$. □

Proposition 4.4. *Let G be a discrete group, $m, r \geq 1$, and let $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$. Suppose an odd $D > 0$ clears its denominators and $\bar{P} = uv$ with $u \in \mathrm{M}_{m\times r}(\mathbb{F}_2[G])$ and $v \in \mathrm{M}_{r\times m}(\mathbb{F}_2[G])$. For every $q = 2^k > 1$ congruent to one modulo D, the matrix A of Lemma 3.2 satisfies*

$$\tau_{r+m}(\log(T_A^*T_A)) = 2(r - \tau_m(P))\log q, \tag{4.1}$$

$$\log \det_{\mathcal{N}(G)}(T_A) = (r - \tau_m(P))\log q. \tag{4.2}$$

It is integral and rationally invertible, its positive-square spectral measure μ_A has support in a compact subinterval of $(0,\infty)$, and

$$\int_{(0,\infty)} \log t \, d\mu_A(t) = \frac{2(r - \tau_m(P))\log q}{r+m}. \tag{4.3}$$

Proof. **Step 1.** Integrality of A and the inverse $A^{-1} \in \mathrm{M}_{r+m}(\mathbb{Q}[G])$ are Lemma 3.2.

Step 2 (reduction to B). By (3.3) and (1.2), the operators $T_{M(x)} = T_{L_x}T_BT_{R_x}$, $x \in [0,1]$, form a path as in Lemma 4.3, with $\Lambda = T_{U/q}$, $\Gamma = T_{V/q}$ and T_B invertible (inverse $T_{B^{-1}}$). Its endpoints are $T_{M(0)} = T_B$ and $T_{M(1)} = T_A$. Lemma 4.3 gives

$$\tau_{r+m}(\log(T_A^*T_A)) = \tau_{r+m}(\log(T_B^*T_B)).$$

Step 3 (the value at B). Since $P^* = P$, we have $J^* = J$ and $B^* = B$. Using $P(I_m - P) = 0$,

$$J^2 = q^{-2}P + (I_m - P), \qquad B^*B = \operatorname{diag}(q^2I_r,\ q^{-2}P + (I_m - P)).$$

The operators $\Pi_1 = \operatorname{diag}(I_r, 0)$, $\Pi_2 = \operatorname{diag}(0, P)$, $\Pi_3 = \operatorname{diag}(0, I_m - P)$ are orthogonal projections with $\Pi_a\Pi_b = 0$ for $a \neq b$ and $\Pi_1 + \Pi_2 + \Pi_3 = I$, and $B^*B = q^2\Pi_1 + q^{-2}\Pi_2 + \Pi_3$. Hence $(B^*B)^j = q^{2j}\Pi_1 + q^{-2j}\Pi_2 + \Pi_3$ for $j \geq 0$, and so

$$f(B^*B) = f(q^2)\Pi_1 + f(q^{-2})\Pi_2 + f(1)\Pi_3 \qquad \text{for every polynomial } f.$$

Taking $f(t) = (t-q^2)(t-q^{-2})(t-1)$ gives $f(B^*B) = 0$, so the spectrum of B^*B lies in $\{q^2, q^{-2}, 1\}$. Let f_0 be a polynomial that agrees with log at these three points. Functional calculus depends only on the values of the function on the spectrum, so $\log(B^*B) = f_0(B^*B)$, that is,

$$\log(B^*B) = 2\log q\,\Pi_1 - 2\log q\,\Pi_2 + 0\cdot\Pi_3 = 2\log q\ \operatorname{diag}(I_r, -P).$$

Its trace is $2\log q\,(r - \tau_m(P))$. With Step 2 this proves (4.1). Half of it is (4.2), by the definition (1.5).

Step 4 (spectral bounds). Since T_A has the bounded inverse $T_{A^{-1}}$,

$$\|T_A^{-1}\|^{-2} I \le T_A^* T_A \le \|T_A\|^2 I,$$

so μ_A is supported in the compact interval $[\|T_A^{-1}\|^{-2}, \|T_A\|^2] \subset (0,\infty)$ and log is bounded there. The identity $\tau_n(f(T_A^*T_A)) = n\int f\,d\mu_A$ of Section 1.1, with $n = r+m$ and $f = \log$, turns (4.1) into (4.3). □

5 Assembly and exact determinant

Proof of Theorem 1.1. **Step 1.** Take K_2, H_2, a_2, b_2 from Theorem 1.2 and put $d = [K_2 : \mathbb{F}_2] \ge 1$. Lemma 2.1 gives $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$ with $XY = I_d \ne YX$.

Step 2. Proposition 2.2 gives the group $G_D = (\bigoplus_{H_2} C_3) \rtimes H_2$, a finite set $S \subseteq H_2 \setminus \{1\}$, and an orthogonal projection $P \in \mathrm{M}_{d+1}(\mathbb{Z}[1/3][G_D])$ with

$$\tau_{d+1}(P) = d + \theta, \qquad \theta = \frac{1}{3}\left(\frac{2}{3}\right)^{|S|} > 0, \qquad \bar{P} = uv \text{ through } d \text{ columns},$$

whose coefficients are cleared by the odd integer $D = 3^{1+|S|}$.

Step 3. Choose $k \ge 1$ with $q = 2^k \equiv 1 \pmod D$. Proposition 4.4 with $G = G_D$, $r = d$, $m = d+1$ produces $A \in \mathrm{M}_{2d+1}(\mathbb{Z}[G_D])$ with an inverse over $\mathbb{Q}[G_D]$ and

$$\log \det_{\mathcal{N}(G_D)}(T_A) = \big(d - (d+\theta)\big)\log q = -\theta \log q.$$

Since $q > 1$ and $\theta > 0$,

$$0 < \det_{\mathcal{N}(G_D)}(T_A) = q^{-\theta} < 1. \tag{5.1}$$

Step 4. With $n = 2d+1$, Equation (4.3) reads

$$\int_{(0,\infty)} \log t\, d\mu_A(t) = -\frac{2\theta \log q}{2d+1} < 0, \tag{5.2}$$

and it is finite by the spectral bounds in Proposition 4.4. This contradicts (1.6).

Step 5 (finite generation). G_D is generated by H_2 and the elements z_h, $h \in H_2$. By (2.2), $z_h = h z_1 h^{-1}$. So a finite generating set of H_2 together with z_1 generates G_D. □

The group G_D differs from the companion's group H_2 and contains the cyclic coordinates of order three. The matrix A has a finite negative logarithmic determinant and μ_A has no mass at zero: the lower bound (1.6) fails, while the logarithmic integral is still finite.

6 Two remarks outside the proof

Remark 6.1 (Right multiplication on row vectors). Some formulations let a matrix act by right multiplication on row vectors. The two conventions are related as follows.

Notation. For $\xi \in \ell^2(G)$ and $g \in G$ let $\xi \cdot g$ be the image of ξ under the unitary $\delta_h \mapsto \delta_{hg}$, and for $\beta \in \mathbb{C}[G]$ put $\xi \cdot \beta = \sum_g \beta_g \, (\xi \cdot g)$; this is right multiplication by β. An element $\xi = (\xi_1, \dots, \xi_n)$ of $\ell^2(G)^n$ is an n-tuple; "row" or "column" only records which product formula is applied. Right multiplication of the row ξ by a matrix $A' \in \mathrm{M}_n(\mathbb{C}[G])$ is $(\xi A')_i = \sum_j \xi_j \cdot A'_{ji}$.

Comparison. Let ι be the unitary on $\ell^2(G)$ with $\iota\delta_g = \delta_{g^{-1}}$; then $\iota^2 = I$. For $g, h \in G$,

$$\iota\lambda_g\iota\,\delta_h = \iota\lambda_g\delta_{h^{-1}} = \iota\delta_{gh^{-1}} = \delta_{hg^{-1}} = \delta_h \cdot g^{-1}.$$

Hence $\iota\lambda(\alpha)\iota\,\xi = \sum_g \alpha_g \, (\xi \cdot g^{-1})$. If α has real coefficients, in particular if $\alpha \in \mathbb{Z}[G]$, then $\alpha^* = \sum_g \alpha_g \, g^{-1}$ and this is $\xi \cdot \alpha^*$. Applying ι in each component of $\ell^2(G)^n$ (still written ι), (1.1) gives, for $A \in \mathrm{M}_n(\mathbb{Z}[G])$,

$$(\iota T_A \iota\,\xi)_i = \sum_j \xi_j \cdot (A_{ij})^* = \sum_j \xi_j \cdot (A^*)_{ji} = (\xi A^*)_i,$$

that is, $\iota T_A \iota$ is right multiplication of the row ξ by the integral matrix A^*. Call this operator r_{A^*}. Then $r_{A^*}^* r_{A^*} = \iota\, T_A^* T_A\, \iota$, and $\iota\epsilon_i = \epsilon_i$. So the spectral measure of $r_{A^*}^* r_{A^*}$ at the vectors ϵ_i equals μ_A. In the row-vector convention the counterexample is therefore the integral matrix A^*, with the same measure and the same determinant. This remark is not used in the proofs.

Remark 6.2 (G_D is not sofic). The source states that applying the sofic determinant bound [3, Theorem 5] to the positive integral matrix A^*A proves that G_D is not sofic. This is an external consequence, not an ingredient in the construction of A, and it depends on a cited theorem whose wording the source does not quote and which this rewrite has not checked against the reference.

What can be checked here is the computation. The operator $T_{A^*A} = T_A^* T_A$ is positive and invertible, and $\log\big((T_A^* T_A)^2\big) = 2\log(T_A^* T_A)$, so by (1.5) and (4.1)

$$\log \det_{\mathcal{N}(G_D)}(T_{A^*A}) = \tau_n\big(\log(T_A^* T_A)\big) = -2\theta \log q < 0.$$

Thus the integral matrix $\Delta = A^*A$, whose operator is positive and invertible, has determinant $q^{-2\theta} < 1$. For the source's conclusion it suffices that the cited theorem yield $\tau_n(\log T_\Delta) \geq 0$ for every sofic group G and every $\Delta \in \mathrm{M}_n(\mathbb{Z}[G])$ with T_Δ positive and invertible. This is the form in which the source invokes it; a reader who wants the non-soficity statement must compare this with the reference.

Verification checklist

Scope. Every step that is *proved* in this document (the conventions of Section 1.1 and Sections 2–5) was checked while writing, and none raised a doubt; the single "Unverified step" box in this document is the one on the cited input, Theorem 1.2, which is not among the proved steps. Three things are *cited*, not proved, and were not checked against their references: Theorem 1.2, on which Theorem 1.1 depends; the identification of (1.6) with [7, Conjecture 13.2], on which only the last sentence of Theorem 1.1 depends; and the sofic theorem in Remark 6.2, on which nothing depends.

Load-bearing points, in suggested order of checking.

1. **The external input.** Theorem 1.2 is cited, not proved; everything is conditional on it (Remark 1.3). It must be checked in the companion [8].
2. **Convention.** T_A is left multiplication on column vectors (1.1), $A \mapsto T_A$ is a $*$-homomorphism (1.2), and the trace is the identity coefficient (1.4). These identify the group-ring algebra with the operator statement in Theorem 1.1.
3. **Separation.** $psp = 0$ for $s \in S$ (Proposition 2.2, Step 3): the factor $1 - p_s$ of p meets the factor p_s of sps^{-1}.
4. **Trace gap.** $[1]p = \theta$ (Proposition 2.2, Step 4), using that the labels $\{1\} \cup S$ are distinct.
5. **Binary factorization.** $uv = \mathrm{diag}(I_d, \bar{p})$ (Proposition 2.2, Steps 5–6), using $XC = 0$, $CY = 0$, $C^2 = C$ from Lemma 2.1 and (2.6).
6. **Lifting.** $W^k = 0$ and $QZ = e$ in Lemma 3.1, Steps 3–4.
7. **Integrality of A.** Lemma 3.2, Step 1: this is where $q \equiv 1 \pmod{D}$ and $UV \equiv e \pmod{q}$ are used.
8. **Block algebra.** $L_x B R_x$ equals the matrix in (3.3), and $M(1) = A$ (Lemma 3.2, Step 3).
9. **Shear invariance.** The derivative formula (Lemma 4.2, Steps 3–4, including the termwise differentiation) and $\tau(M'M^{-1}) = 0$ (Lemma 4.3).
10. **Evaluation.** $\log(B^*B) = 2 \log q \, \mathrm{diag}(I_r, -P)$ (Proposition 4.4, Step 3) and the substitution $r = d$, $\tau_{d+1}(P) = d + \theta$ (Section 5, Step 3).

References

[1] Pierre de la Harpe and Georges Skandalis. Déterminant associé à une trace sur une algèbre de Banach. *Annales de l'Institut Fourier*, 34(1):241–260, 1984.

[2] Warren Dicks and Thomas Schick. The spectral measure of certain elements of the complex group ring of a wreath product. *Geometriae Dedicata*, 93:121–137, 2002.

[3] Gábor Elek and Endre Szabó. Hyperlinearity, essentially free actions and L^2-invariants. The sofic property. *Mathematische Annalen*, 332:421–441, 2005.

[4] Bent Fuglede and Richard V. Kadison. Determinant theory in finite factors. *Annals of Mathematics*, 55(3):520–530, 1952.

[5] Holger Kammeyer. A counterexample to the determinant approximation conjecture. arXiv:2609.15567v1, 2026. Version 1, September 14, 2026.

[6] Franz Lehner, Markus Neuhauser, and Wolfgang Woess. On the spectrum of lamplighter groups and percolation clusters. *Mathematische Annalen*, 342(1):69–89, 2008.

[7] Wolfgang Lück. *L^2-Invariants: Theory and Applications to Geometry and K-Theory*, volume 44 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge*. Springer, 2002.

[8] OpenAI. A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Characteristic Two. OpenAI Math Release preprint OAI:A-Counterexample-to-Kaplanskys-Direct-Finiteness-Conjecture-in-Characteristic-Two-September-23-2026, 2026. Theorem 1.1 and Section 6.

[9] Thomas Schick. L^2-determinant class and approximation of L^2-Betti numbers. *Transactions of the American Mathematical Society*, 353(8):3247–3265, 2001. With the appended erratum in arXiv:math/9807032v2.